

SeaLink: Integrated Seaport & Cargo Management System

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Group 10 — Tutorial 1

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1 Project Scope (Problem Statement)

The **Port Authority of Gujarat** is automating its International Commercial Seaport operations with the help of a centralized **Database Management System (DBMS)**. Presently, due to the lack of coordination at different levels of stakeholder groups, congestion, safety hazards, and loss of revenue are encountered. **SeaLink** addresses these issues by integrating the following four core stakeholders involved in port operations: **Shipping Agents, Port Operations Managers, Customs Authorities, and System Administrators**.

The process begins with the **Shipping Agent**, who represents the vessel owner. The agent accesses the system to register a vessel and book a docking slot for a particular date. Essential information such as **vessel size** and **type of cargo** must be provided. Additionally, the agent may upload a detailed **cargo manifest** in advance of the ship's arrival, allowing the port to plan activities efficiently.

The system enables the **Port Operations Manager** to carry out efficient **resource allocation**. Docking spots are assigned based on vessel size and berth availability. Critical equipment such as cranes and tugboats are continuously monitored to ensure they are not scheduled when maintenance is due. **Safety** is a priority, and in cases of high winds or adverse weather conditions, operations are automatically shut down to prevent accidents. When port capacity is limited, vessels carrying essential or perishable goods are **prioritized** over those with regular cargo.

Once a vessel docks and unloading begins, the system **monitors the movement** of every container. It enforces **strict safety rules** to ensure that dangerous goods are not stored near incompatible materials. It also **optimizes container stacking** in the yard

so that containers scheduled for early departure are easily accessible, thereby minimizing unnecessary crane movements.

Before cargo can leave the port by truck, rail, or another vessel, approval from the **Customs Authority** is required. The system enforces this through a **digital clearance mechanism**: no exit pass can be generated unless a Customs Officer has marked the cargo as cleared. Finally, SeaLink automatically calculates all applicable fees, including docking charges and storage fines, ensuring accurate and transparent **billing**.

2 Detailed Functionality

2.1 Shipping Agent (Vessel Representative)

- **Vessel Registration:** Registers ships and uploads the detailed list of incoming goods (Cargo Manifest).
- **Berth Booking:** Requests docking slots based on vessel size and views real-time estimates for when the ship will be unloaded.
- **Financial Dashboard:** Views and pays invoices generated for docking fees, pilotage services, and utility usage.
- **Live Tracking:** Monitors the exact status of their vessel (e.g., Waiting at Anchor, Currently Unloading) and their cargo.

2.2 Port Operations Manager (Service Provider)

- **Resource Allocation:** Assigns Docking Spots, Tugboats, and Harbor Pilots based on availability and vessel requirements.
- **Asset Health:** Monitors a maintenance scheduler that tracks engine hours for cranes and tugs to prevent breakdowns.
- **Safety Protocols:** Manages emergency states (like closing the port during storms) and oversees the handling of dangerous goods.
- **Yard Optimization:** Oversees the logic that organizes the storage yard to maximize space and speed up truck loading.

2.3 Customs Officer (Regulatory Authority)

- **Clearance Lock:** Uses a mandatory validation tool where cargo is marked as “Cleared” or “Held” for inspection.
- **Gate Control:** Ensures that no cargo is issued a “Gate Pass” or loaded onto a train without prior clearance.
- **Inspection:** Verifies that declared dangerous goods comply with international safety standards.

2.4 System Administrator

- **Master Data:** Manages the physical layout of the port in the database (adding new docking spots or storage blocks).
- **User Management:** Verifies the credentials and licenses of new Shipping Agents before giving them access.
- **Audit Logs:** Reviews security logs to track all critical changes and ensure system integrity.

3 Supported Queries

1. **Operational:** “Find all containers containing dangerous goods currently stored in 'Block A' that have not yet been cleared by Customs.”
2. **Asset Health:** “List all Cranes that have exceeded their safe operating hours and require immediate service.”
3. **Logistical:** “Identify all containers scheduled to leave via 'Rail' today that are currently buried at the bottom of a stack.”
4. **Financial:** “Calculate the total storage fines collected from a specific Shipping Line in the last quarter.”
5. **Efficiency Analysis:** “Calculate the average 'Turnaround Time' (Arrival to Departure) for vessels this month compared to the previous month to measure port efficiency.”
6. **Traffic Monitoring:** “Identify the peak congestion hours at the Gate Complex by counting the number of trucks entering between 8:00 AM and 8:00 PM.”

4 Contribution

With combined discussions and meets, we all contributed equally in completing the deadline 1.